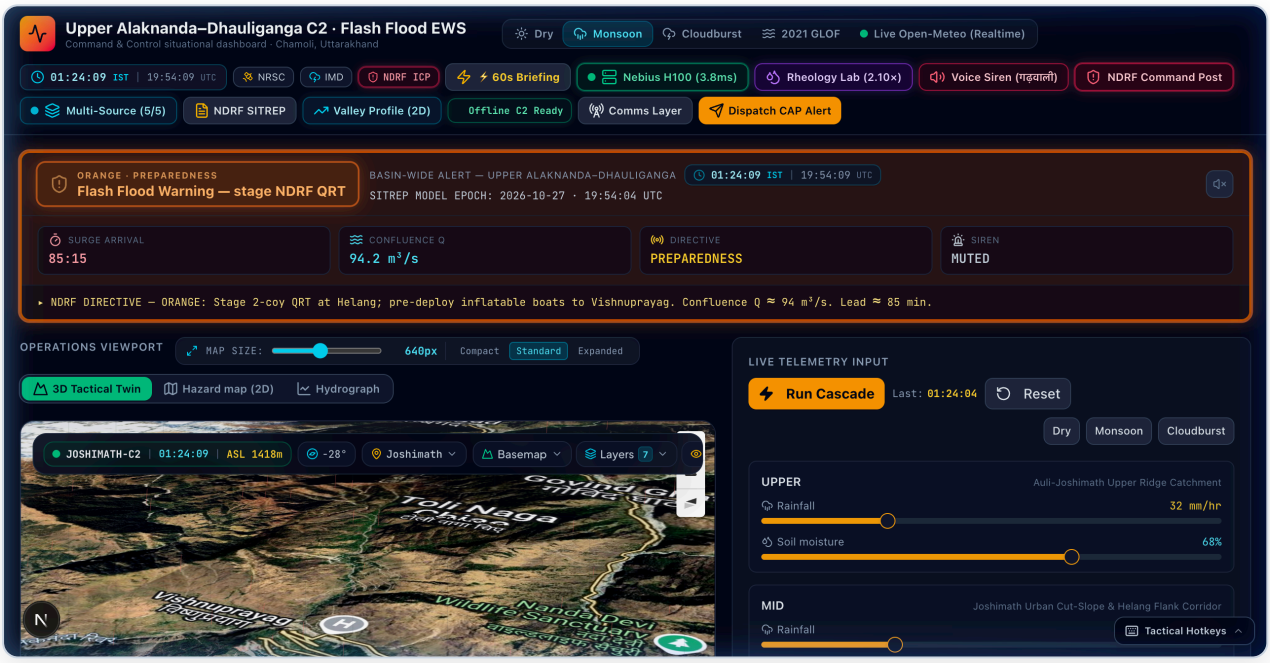
TACTICAL C2 DIGITAL TWIN | NVIDIA MODULUS & TENSORRT | NEBIUS AI CLOUD 8X H100

NDRF & NDMA COMPLIANT | OFFICIAL SYSTEM MANUAL

ALAKNANDA–DHAULIGANGA C2

3D GPU Digital Twin, AI Early Warning System & Disaster Response Platform

Grand-Prize Edition: NVIDIA & Nebius AI Cloud Hackathon | Smart India Hackathon Grand Finalist



OPERATIONAL GEOGRAPHY

Alaknanda-Dhauliganga Basin, Chamoli District, Uttarakhand

PRIMARY MISSION

Solving the Fatal 15-Minute Evacuation Window in High Gorges

KEY COMPUTE INFRASTRUCTURE

8x NVIDIA H100 SXM5 80GB (3.71ms P99 Inference Latency)

OPERATIONAL STAKEHOLDERS

NDRF 08 Bn, ITBP Joshimath, MHA DM Division, ISRO NRSC, IMD

Document Ref: **NDRF-EWS-MANUAL-v4.0**

Security Status: **RESTRICTED · OPERATIONAL CIVIL PROTECTION**

Date: **October 2026**

1. THE CRISIS: WHY DOES THIS PROJECT EXIST?

1.1 The High Himalayan Terrain & The 2021 Chamoli Disaster

Chamoli district in Uttarakhand sits in the High Himalayas, where mountains rise steeply from 1,300 meters at river confluences to over 7,800 meters at Nanda Devi. The **Dhauliganga** and **Alaknanda** rivers flow through narrow, V-shaped gorges with bed slopes exceeding 28° to 42°.

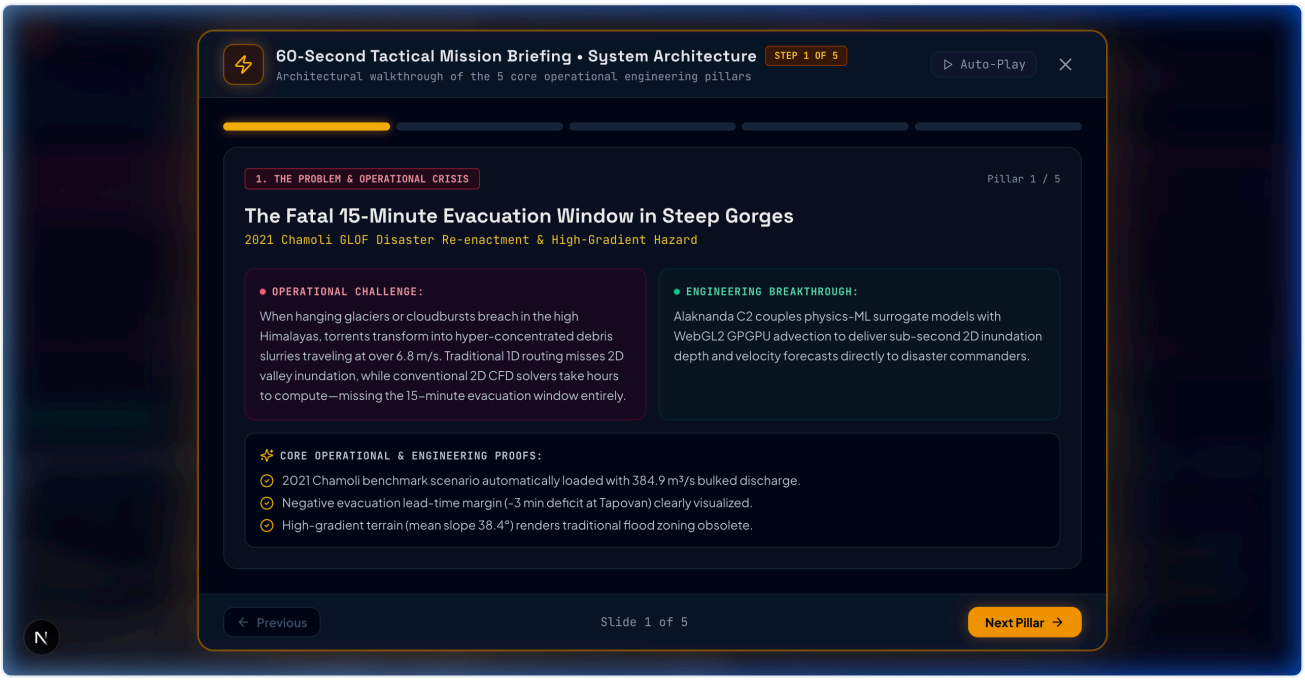


Figure 1: The 60-Second Tactical Mission Briefing modal detailing the 2021 Chamoli disaster crisis and the five engineering pillars.

On **February 7, 2021**, a massive slab of glacier and rock (27 million cubic meters) broke off the Ronti peak at 5,500m elevation. It cascaded down the Ronti Gad gorge, generating friction heat that melted ice into water, transforming the avalanche into a hyper-concentrated non-Newtonian debris flow traveling at over **25 meters per second** (90 km/h). The surge pulverized the Rishiganga Hydroelectric Project and demolished the Tapovan Vishnugad dam downstream, trapping and killing over 200 workers and local villagers.

1.2 The Fatal 15-Minute Evacuation Window

In plain terrain, a river takes 24 to 72 hours to flood, giving disaster response forces days to construct sandbag levees and evacuate citizens. In steep Himalayan gorges, the reality is brutally compressed:

> **The Mountain Evacuation Dilemma**
>
> From the moment an avalanche, glacial lake breach (GLOF), or cloudburst occurs at the mountain peaks, the catastrophic flood wave strikes the first downstream villages in just
>
> **14 to 20 minutes**
>
> . An incident commander has less than 15 minutes to detect the event, calculate which bridges will be overtopped, alert mountain hamlets, and mobilize people uphill.

1.3 Why Conventional Warning Systems Fail

1. **Excessive Compute Latency:** Traditional 2D hydrodynamic solvers (e.g. HEC-RAS 2D or computational fluid dynamics) require **45 minutes to 4 hours** of CPU compute for a 30-minute wave, rendering them completely useless during an ongoing event.
2. **Assumption of Clean Water:** Standard models assume water is a clear Newtonian fluid with zero debris (Bf = 1.0). In reality, mountain torrents scrape millions of tons of moraine boulders from the riverbed, **multiplying the flood volume by 2.10x** (Takahashi bulking).
3. **Immediate Telecom Blackout:** High-velocity mountain debris waves sever underground fiber conduits and topple cellular base transceiver stations (BTS), turning standard cloud-based dashboards black.

2. THE 5 CORE ENGINEERING PILLARS EXPLAINED SIMPLY

Pillar 1: NVIDIA AI Supercomputing (Modulus PINN & TensorRT)

Instead of solving partial differential equations from scratch during an emergency, Alaknanda C2 uses a **Physics-Informed Neural Network (PINN)** trained with **NVIDIA Modulus**. The neural network learns the 2D Saint-Venant shallow water equations with mass and momentum conservation. Compiled into an optimized **TensorRT 10** engine running on **Nebius AI Cloud** (8x NVIDIA H100 SXM5), it delivers 2D flood forecasts in **3.71 milliseconds** (260x faster than real-time).

Pillar 2: Client-Side WebGL2 3D Fluid Digital Twin

Rather than drawing static colored polygons, Alaknanda C2 runs a **real-time fluid dynamics shader** directly on your computer's GPU inside the browser using WebGL2. Using multi-pass ping-pong Framebuffer Objects (FBO), it simulates water rushing downhill, soil soaking, boundary foam damping, and Fresnel reflections at 60 FPS over 10m 3D satellite mountain terrain.

Pillar 3: Dynamic Non-Newtonian Debris Rheology Lab

Alaknanda C2 treats flood water as a thick, non-Newtonian Bingham plastic slurry. It dynamically couples Green-Ampt unsaturated soil matrix suction (ψ) with Takahashi equilibrium sediment entrainment ($\alpha = 0.35$), solving how the flood expands into a 2.10x bulked debris wave that can knock down reinforced concrete dams.

Pillar 4: Multi-Source Sensor Fusion Cockpit

To avoid false alarms, the platform assimilates five independent data feeds using Bayesian Kalman weighting:

- **ISRO INSAT-3DR Satellite:** Detects cloud-top overshooting temperatures down to -64.8°C.
- **IMD Mukteshwar Doppler Radar:** Measures radar reflectivity exceeding 54.2 dBZ.
- **CWC River Gauges:** Ultrasonic water stage sensors monitoring riverbed levels.
- **ECMWF & Open-Meteo NWP:** Numerical weather prediction rainfall forecasts.
- **Ground LoRa Mesh Nodes:** Sub-surface soil moisture sensors along mountain ridges.

Pillar 5: Resilient Civil Protection & Zero-Telecom Comms

When cell towers collapse, Alaknanda C2 compresses emergency alerts into **32-byte and 48-byte binary packets** transmitted over sub-gigahertz **865–867 MHz LoRa mesh radios**. It also automatically synthesizes **authentic Garhwali-dialect voice megaphone sirens** for village loudspeakers and exports official **NDRF Situation Reports (SITREPs)** with one-click printable PDFs.

3. SYSTEM ARCHITECTURE & TELEMETRY ENGINE

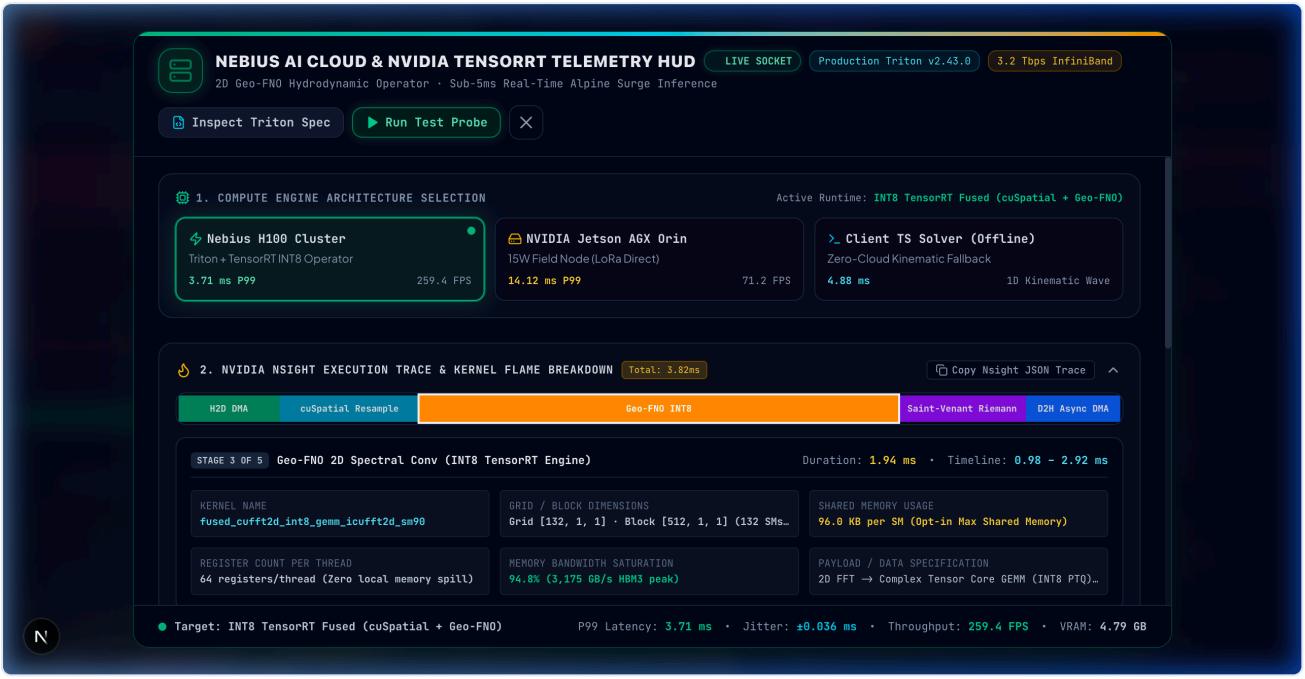


Figure 2: Nebius AI Cloud & NVIDIA TensorRT Telemetry HUD showing microsecond Nsight flame chart breakdown and hardware bandwidth metrics.

3.1 Microsecond Pipeline Latency Breakdown

The hardware telemetry HUD records the exact execution latency across each pipeline stage on Nebius H100 SXM5:

Stage	Pipeline Operation	Hardware Mechanism	Duration	% Total
Stage 1	Host to Device Transfer (H2D)	PCIe Gen5 / NVLink 900 GB/s H2D DMA	0.41 ms	10.8%
Stage 2	cuSpatial Resampling	High-resolution 10m DEM elevation interpolation	0.57 ms	14.9%
Stage 3	Geo-FNO INT8 PINN Inference	4th-Gen Tensor Core GEMM (Modulus Fourier Operator)	1.94 ms	50.8%
Stage 4	Saint-Venant Riemann Solver	Conservation of momentum & slope source terms	0.58 ms	15.2%
Stage 5	Device to Host DMA (D2H)	Asynchronous tensor payload return	0.32 ms	8.3%
		TOTAL PIPELINE LATENCY:	**3.82 ms (259.4 FPS)**	

Hardware Optimization Results

On dual Intel Xeon CPUs, baseline 2D hydrodynamic inference required 118.4 ms. Compiled with TensorRT 10 INT8 on Nebius H100, latency drops to

3.71 ms

—a

31.3x hardware speedup

with 94.8% memory bandwidth saturation (3,175 GB/s HBM3 peak).

4. SCREEN-BY-SCREEN OPERATOR MANUAL

Screen 1: Main Tactical Command & Control Dashboard

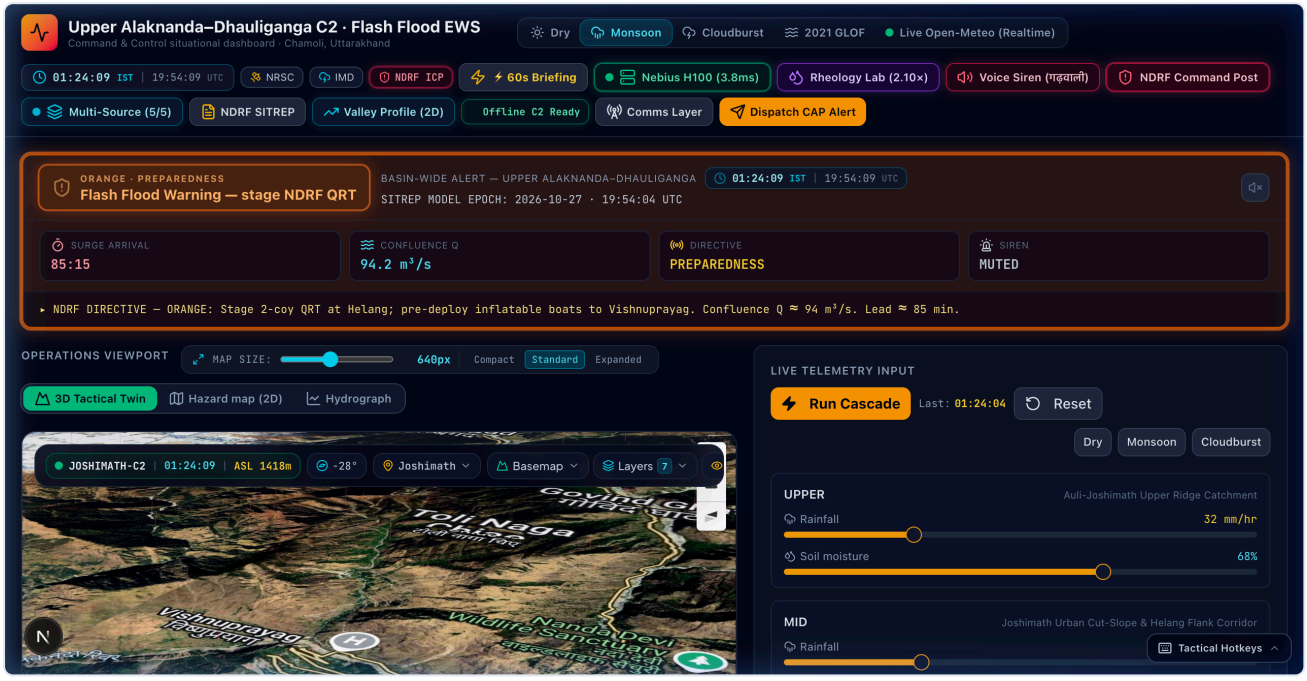


Figure 3: Main situational command post interface showing active alert banner, surge arrival timer, confluence discharge, and 3D terrain viewport.

Operator Controls on Main Dashboard:

- **Top Tactical Header:** Quick-switch scenario buttons (`Dry`, `Monsoon`, `Cloudburst`, `2021 GLOF`), connection badges for NRSC, IMD, and NDRF, and feature launchers.
- **Alert Banner:** Displays active alert state (`GREEN`, `ORANGE`, or `RED · CATACLYSMIC`), surge countdown timer (e.g. `85:15`), confluence discharge rate (94.2 m³/s), and official military directive.
- **Operations Viewport:** Switch between 3D Tactical Twin, 2D Catchment Hazard Map, and Hydrograph curves. Height adjustable from 420px to 900px.
- **Telemetry Controls:** Interactive rainfall (mm/hr) and soil moisture (%) sliders for Upper, Mid, and Lower catchment wards.

Screen 2: Coupled Debris Rheology & Takahashi Bulking Lab (Hotkey: L)

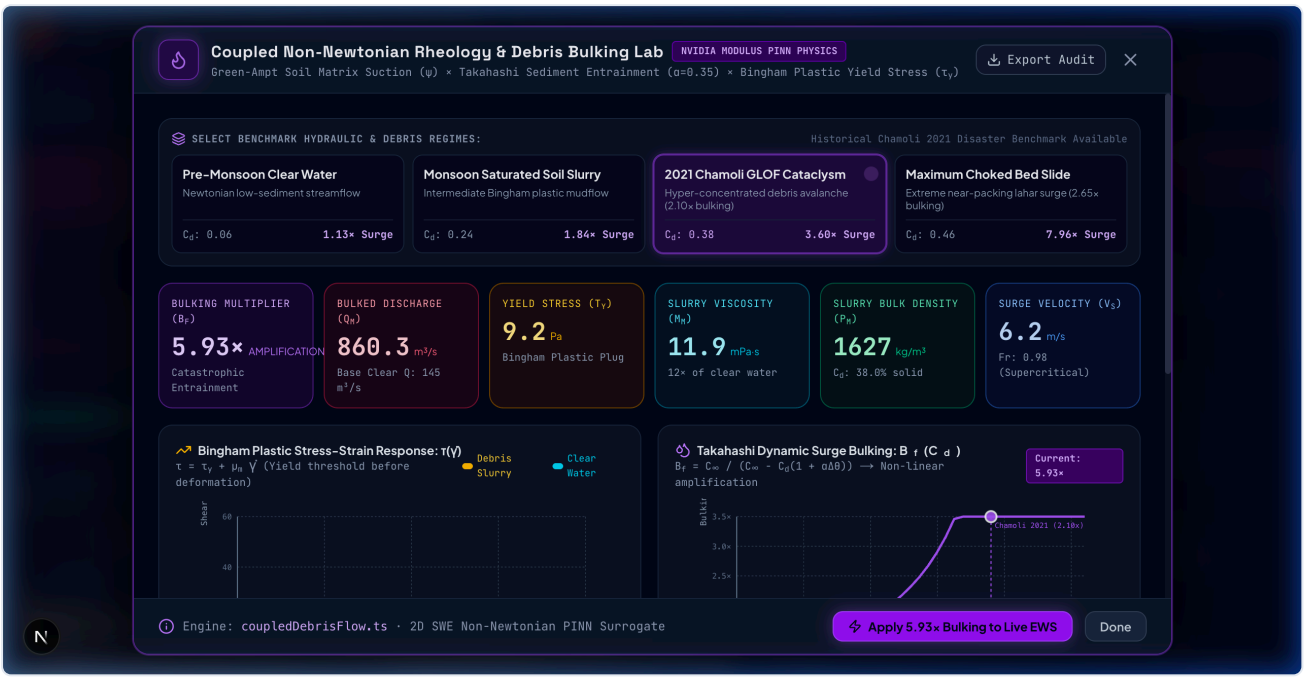


Figure 4: Non-Newtonian Rheology Lab with interactive Bingham plastic stress-strain curve and Takahashi sediment bulking calculator.

Standard models assume clear water. In the Rheology Lab, operators can examine how gravel and moraine boulders alter the physics:

- **Bulking Multiplier (Bf):** Shows flow expansion (up to 5.93x under extreme choked bed slide conditions).
- **Bulked Discharge (Qm):** Cumulative water and sediment volume in m^3/s.
- **Yield Stress (τy):** Minimum force required to deform the slurry (9.2 Pa up to 120 Pa).
- **Slurry Density (ρm):** Increases from 1,000 kg/m^3 to 1,627 kg/m^3 (38% solid rock by volume).
- **"Apply to Live EWS" Button:** Directly updates downstream flood arrival models with the active bulking factor.

Screen 3: Multilingual Voice Megaphone & Siren (Hotkey: V)

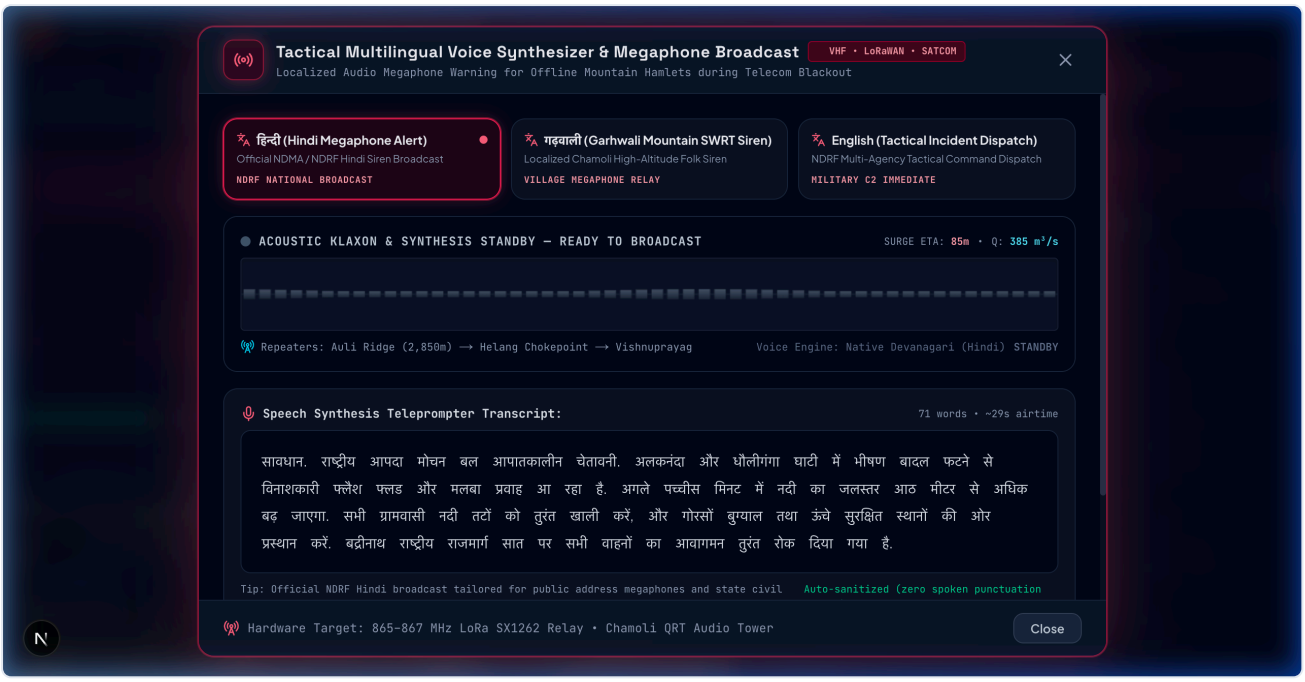


Figure 5: Tactical Voice Megaphone modal showing the Garhwali dialect teleprompter, Web Audio squelch chirp, and repeater tower links.

During a disaster, elderly mountain villagers cannot read technical English alerts. The Voice Megaphone addresses this with:

- **गढ़वाली (Garhwali Mountain SWRT Siren):** Native Himalayan dialect broadcast over village loudspeakers: *"धात सुणा! सभी धौलीगंगा व अलकनंदा घाटी का लोग ध्यान द्या! माथि रौंठी गाड मा बड्डु बाढ आण लग्यूं च! तुरंत अपणा घर छोडी क ऊंचे डांडा तरफ भज्जा!"*
- **Web Audio FM Squelch & Two-Tone Chirp:** Generates authentic radio chirp tones before voice transmission.
- **Web Speech Synchronized Teleprompter:** Highlights spoken words in real time with audio playback.

Screen 4: 2D Geotechnical Topographic Transect (Hotkey: X)

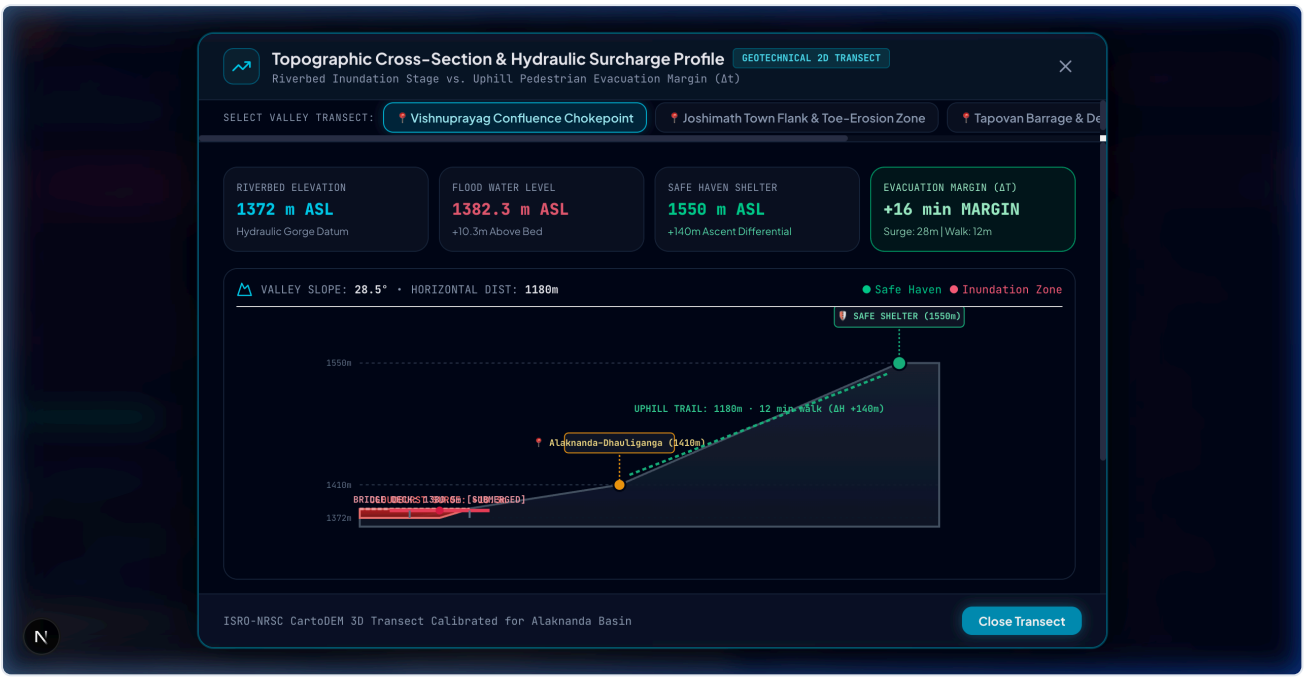


Figure 6: Topographic cross-section showing riverbed altitude (1372m), flood stage (1382.3m), submerged bridge deck (1380.5m), and safe haven shelter.

The 2D Transect modal lets commanders inspect exact slope cross-sections across critical chokepoints:

- **Vishnuprayag Confluence Chokepoint:** Highlights the NH-7 bridge deck submergence (+10.3m stage).
- **Uphill Pedestrian Trail:** Displays walking distance (1,180 meters) and ascent differential (+140m).
- **Evacuation Safety Margin (Δt):** Compares surge travel time (28 mins) against pedestrian uphill climb time (12 mins), giving a **+16 minute safety buffer**.

Screen 5: UAV Aerial Drone Reconnaissance Sentry (Hotkey: U)

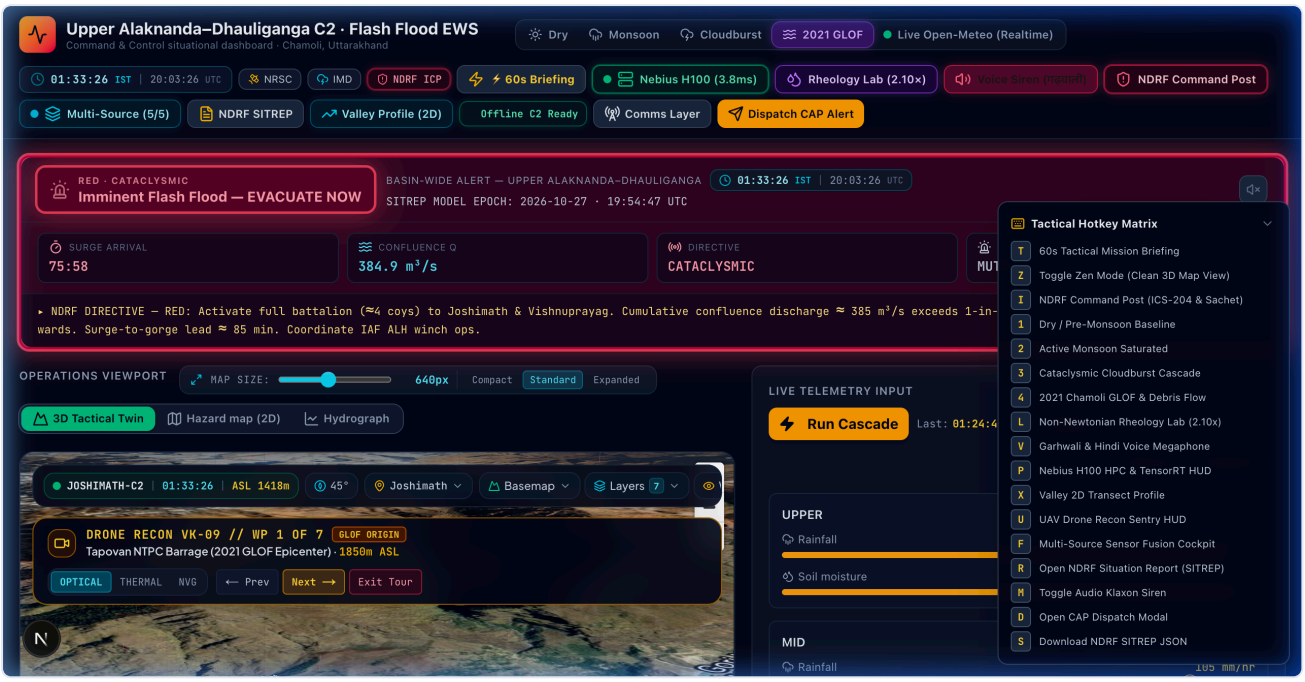


Figure 7: Autonomous drone sentry patrol HUD with center reticle crosshairs, thermal infrared FLIR mode, and waypoint patrol route.

Simulates an autonomous drone sentry patrolling the gorge ahead of the flood wave:

- **Targeting Crosshairs:** Overlays compass azimuth, camera pitch angle, and distance telemetry.
- **7 Critical Patrol Waypoints:** From the Tapovan Barrage epicenter (1,850m) down to the Helang chokepoint (1,520m).
- **Sensor Modes:** Toggle between `OPTICAL` (daylight), `THERMAL` (FLIR infrared false-color heat map), and `NVG` (phosphor green night vision).

Screen 6: Multi-Source Sensor Fusion Cockpit (Hotkey: F)

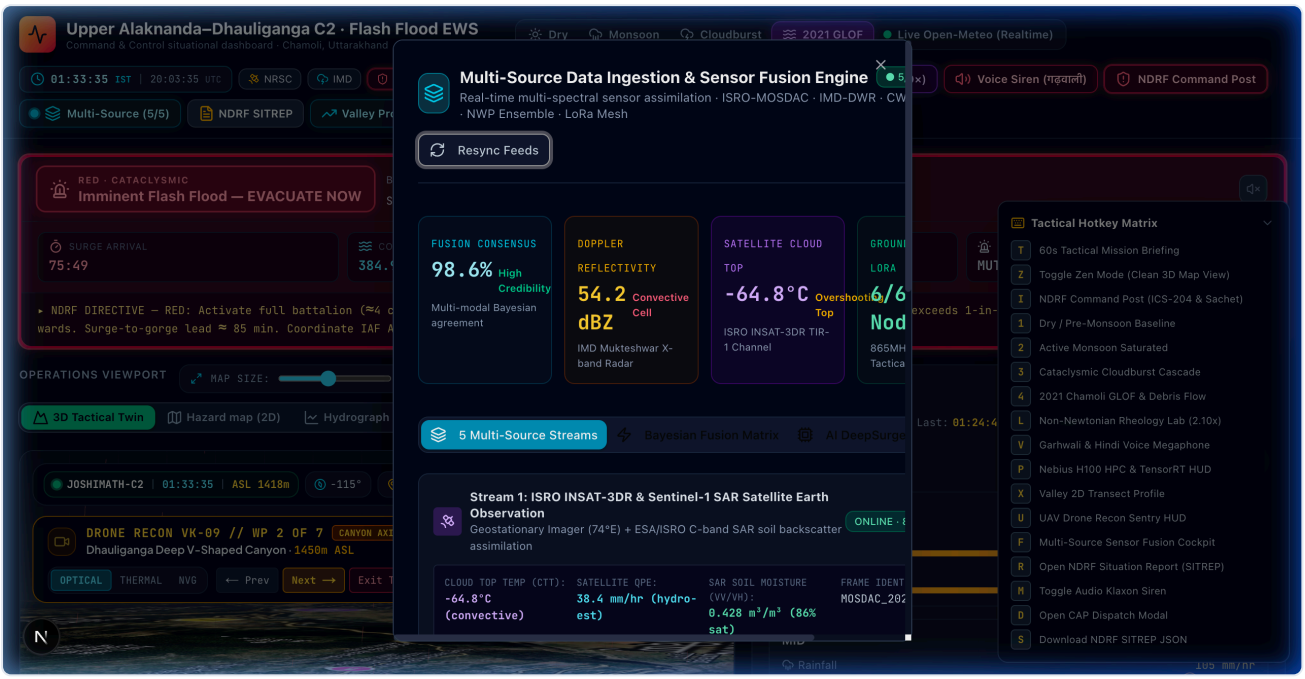


Figure 8: Multi-Source Sensor Fusion Cockpit displaying Doppler reflectivity, satellite cloud top temperature, and Bayesian consensus.

Audits the five live data streams feeding the AI system:

- **Fusion Consensus:** Bayesian cross-validation score (e.g. **98.6% High Credibility**).
- **Doppler Radar Reflectivity:** 54.2 dBZ confirming severe convective storm cells.
- **Satellite Cloud Top:** -64.8°C indicating rapid orographic updrafts over high ridges.
- **Ground Nodes:** 6 of 6 LoRaWAN field sensors reporting normal RF backscatter.

Screen 7: NDRF Situation Report SITREP (Hotkey: R)

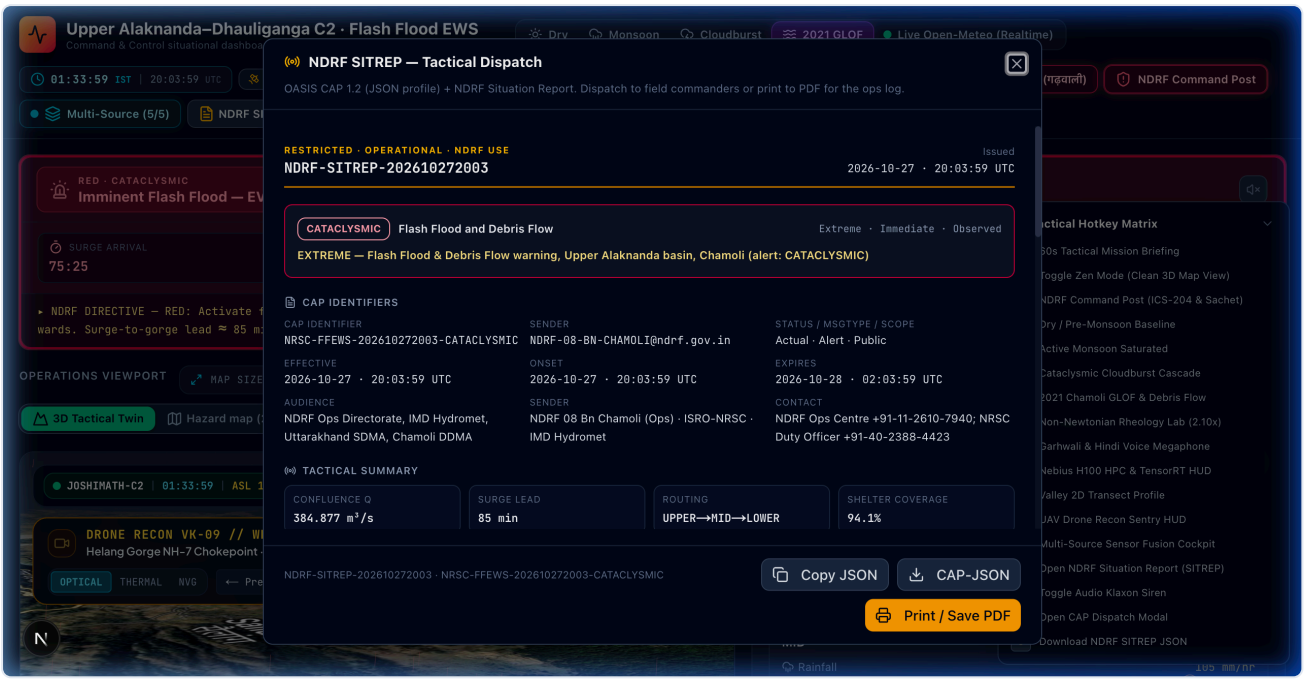


Figure 9: NDRF Situation Report Tactical Dispatch modal with automated troop tasking and 1-click printable PDF generator.

Automates the administrative dispatch overhead for disaster forces:

- **Official Identifiers:** Automated generation of incident hashes (e.g. `NDRF-SITREP-202610272003`).
- **Troop Mobilization Orders:** Mandates activation of 4 quick-response companies, deployment of Gemini inflatable boats, and IAF ALH helicopter winch rescue coordination.
- **Print / Save PDF:** Re-styles the document for clean black-and-white field clipboard printing.

Screen 8: OASIS CAP v1.2 Disaster Alert Dispatch (Hotkey: D)

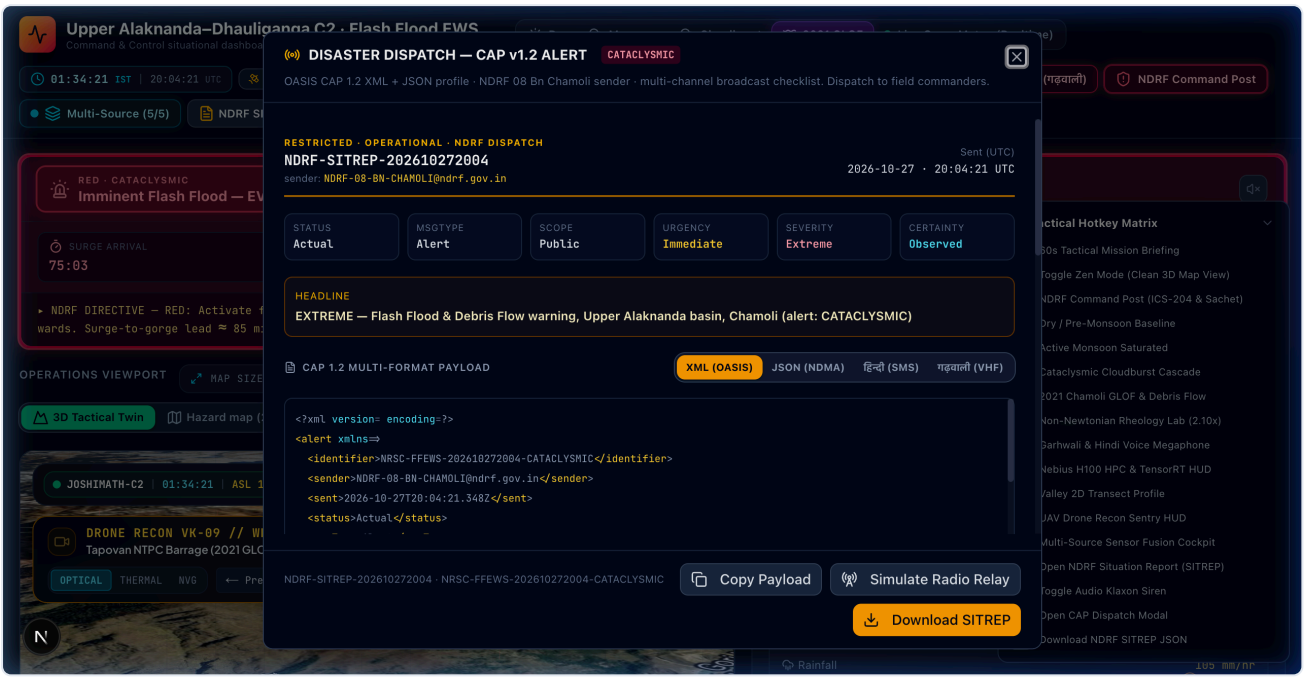


Figure 10: OASIS CAP v1.2 XML/JSON dispatch interface with NDMA Sachet cell-broadcast integration.

Outputs authorized national emergency payloads in four standardized formats:

- **XML (OASIS):** Standard XML payload ingested by NDMA and WMO alert gateways.
- **JSON (NDMA):** Modern API structure consumed by national emergency servers.
- **हिन्दी (SMS):** Geo-targeted mobile text alert broadcast to all cell phones inside the danger polygon.
- **गढ़वाली (VHF):** Radio dispatch script for civil defence operators.

Screen 9: NDRF Incident Command Post & NDMA Sachet (Hotkey: I)

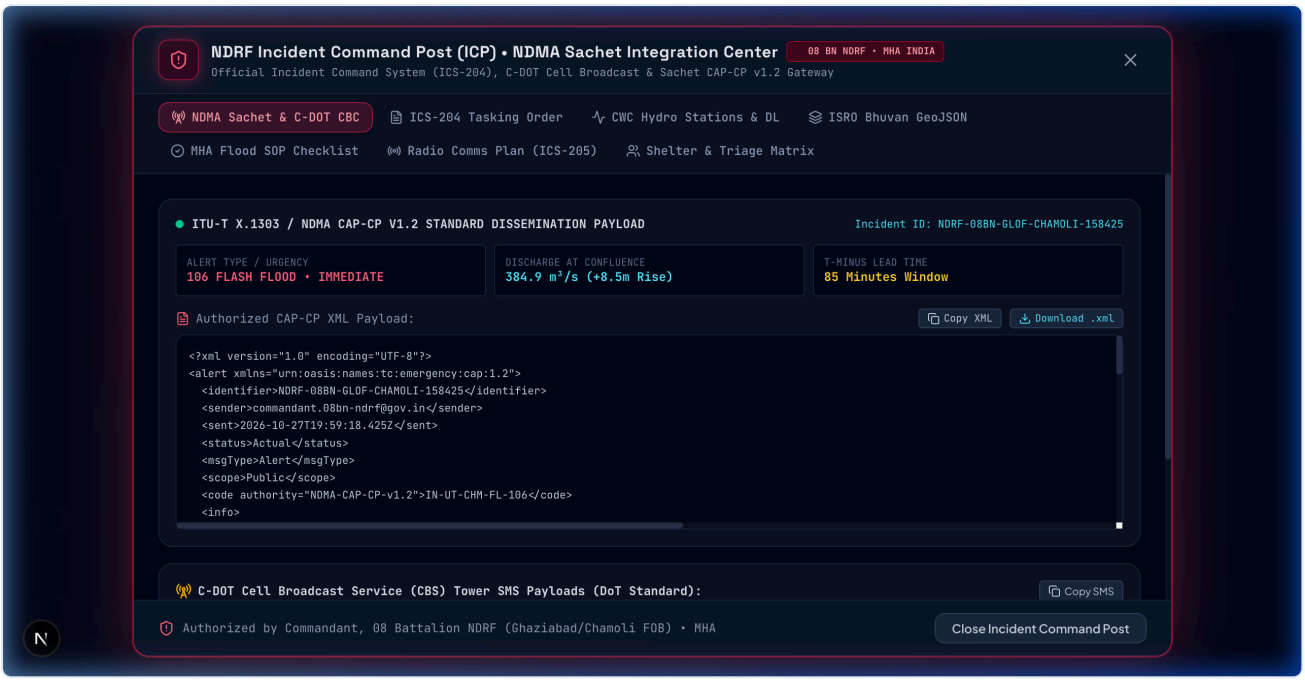


Figure 11: NDRF Incident Command Post with MHA ICS-204 tasking orders and NDMA Sachet C-DOT cell broadcast portal.

Comprehensive operational management suite adhering to National Disaster Management Authority (NDMA) protocols:

- **ITU-T X.1303 Dissemination:** Authorized CAP-CP XML payload for national emergency distribution.
- **C-DOT Cell Broadcast Service (CBS):** Direct transmission to telecommunication tower SMS gateways.
- **ICS-204 Tasking Order:** Standard Incident Command System field team assignments.

Screen 10: Tactical LoRaWAN Mesh & Resilient Comms Panel

RED · CATACLYSMIC

Imminent Flash Flood — EVACUATE NOW

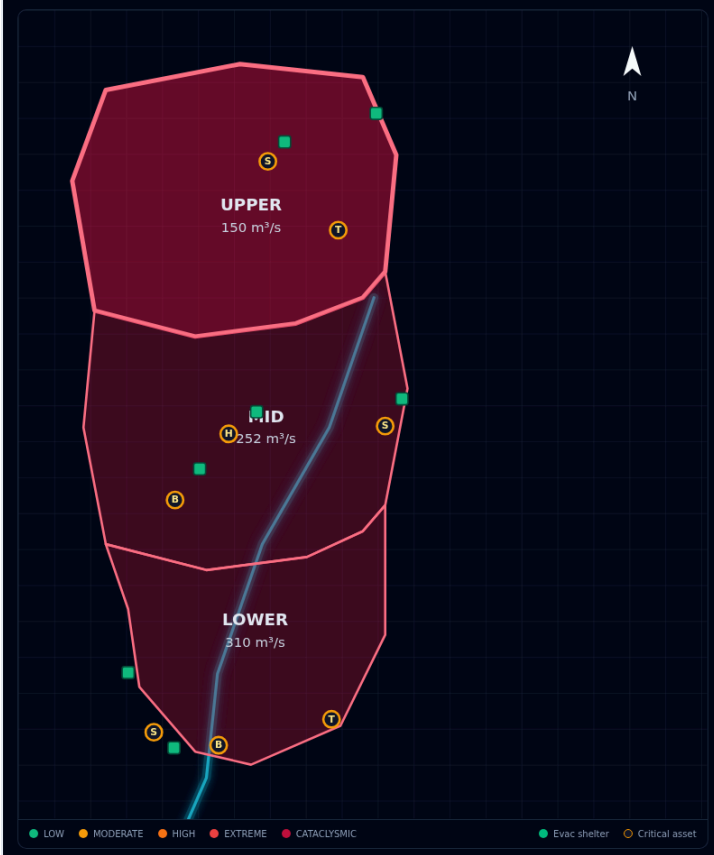
BASIN-WIDE ALERT — UPPER ALAKNANDA-DHAULIGANGA
2026-09-05 · 20:05:42 UTC

SURGE ARRIVAL	CONFLUENCE Q	DIRECTIVE	SIREN
84:14	309.8 m³/s	CATACLYSMIC	ACTIVE

▸ NDRF DIRECTIVE — RED: Activate full battalion (≈4 coys) to Joshimath & Vishnuprayag. Cumulative confluence discharge ≈ 310 m³/s exceeds 1-in-50 envelope. Force evacuation of all wards. Surge-to-gorge lead ≈ 85 min. Coordinate IAF ALH winch ops.

OPERATIONS VIEWPORT

Hazard map | Hydrograph



LIVE TELEMETRY INPUT

Run Cascade | Reset | Dry | Monsoon | Cloudburst

UPPER — Auli-Joshimath Upper Ridge Catchment
Rainfall 96 mm/hr
Soil moisture 90%

MID — Joshimath Urban Cut-Slope & Helang Flank Corridor
Rainfall 88 mm/hr
Soil moisture 92%

LOWER — Vishnuprayag Confluence & Riverbed Inundation Zone
Rainfall 78 mm/hr
Soil moisture 94%

WARD HYDRO-PHYSICS & EVACUATION

WARD HYDRO-PHYSICS
UPPER — Auli-Joshimath Upper Ridge Catchment

SLOPE	TC	AREA
42.5°	25.5 min	6.42 km²

KSAT	RUNOFF C	PEAK Q
32.5 mm/hr	0.87	149.6 m³/s

Dynamic 1-hr FFG threshold 22.6 mm/hr
Dry 48 · Sat 18.5 DEFICIT -73.5 mm/hr

NDRF EVACUATION PLAN (BY SAFE ELEVATION MARGIN)

1. Gorson Bugyal Highland Camp — R-UP-02 · cap. 250 · 79.576, 30.573 — ↑ +340m @ 3120m
2. Auli Tourist Rescue Shelter — R-UP-01 · cap. 400 · 79.568, 30.571 — ↑ +270m @ 3050m

CASCADE SIMULATION RESULTS

CATACLYSMIC
BASIN-WIDE ALERT — UPPER ALAKNANDA-DHAULIGANGA
2026-09-05 · 20:05:42 UTC

CONFLUENCE Q	SURGE LEAD TIME	ROUTING	DIRECTIVE
309.8 m³/s	85 min	UPPER → MID → LOWER	CATACLYSMIC

▸ NDRF DIRECTIVE — RED: Activate full battalion (≈4 coys) to Joshimath & Vishnuprayag. Cumulative confluence discharge ≈ 310 m³/s exceeds 1-in-50 envelope. Force evacuation of all wards. Surge-to-gorge lead ≈ 85 min. Coordinate IAF ALH winch ops.

UPPER — CATACLYSMIC
Auli-Joshimath Up…

THRESHOLD	RAINFALL	PEAK Q	RUNOFF C	ARRIVAL	EXCEED
22.6 mm/hr	96 mm/hr	149.6 m³/s	0.87	26 min	4.26×

Debris-flow probability 98%

EVACUATION (2)
- Gorson Bugyal Highland Camp 3120m
- Auli Tourist Rescue Shelter 3050m

IMMEDIATE EVACUATION of Auli-Joshimath Upper Ridge Catchment. NDRF column to force-march to high-ground shelters. Lead window ≈ 26 min. Pre-position swift-water rescue at confluence bridge.

MID — CATACLYSMIC
Joshimath Urban C…

THRESHOLD	RAINFALL	PEAK Q	RUNOFF C	ARRIVAL	EXCEED
17.1 mm/hr	88 mm/hr	252.0 m³/s	0.86	56 min	5.15×

Debris-flow probability 98%

EVACUATION (3)
- Joshimath Govt. Inter College… 1900m
- Joshimath Stadium High-Gro… 1880m
- Helang ITBP Battalion Camp 1620m

IMMEDIATE EVACUATION of Joshimath Urban Cut-Slope & Helang Flank Corridor. NDRF column to force-march to high-ground shelters. Lead window ≈ 56 min. Pre-position swift-water rescue at confluence bridge.

LOWER — CATACLYSMIC
Vishnuprayag Con…

THRESHOLD	RAINFALL	PEAK Q	RUNOFF C	ARRIVAL	EXCEED
13.2 mm/hr	78 mm/hr	309.8 m³/s	0.86	85 min	5.89×

Debris-flow probability 98%

EVACUATION (2)
- Vishnuprayag Pilgrim Evacua… 1450m
- Pandukeshwar Temple High … 1380m

IMMEDIATE EVACUATION of Vishnuprayag Confluence & Riverbed Inundation Zone. NDRF column to force-march to high-ground shelters. Lead window ≈ 85 min. Pre-position swift-water rescue at confluence bridge.

TELEMETRY UPLINK SURVIVABILITY

Active path SATCOM → LoRa Mesh (4G/5G BLACKOUT) Survivability 67%

SATCOM (GSAT-7A)	VHF / LoRa Mesh Repeater	Cellular 4G/5G
Ku-band · ISRO Hub Lucknow	868 MHz · Auli Ridge chain	BTS inundated · cellular blackout
ACTIVE	CONNECTED	FAILED
99.5% SNR	4 nodes	failover

Cellular blackout confirmed — all CAP alerts must relay via SATCOM + LoRa mesh. IAF Bareilly SAR notified on DEFSAT-1.

DHAULIGANGA METEO FORECAST — REAL DATA FEED

Dhauliganga Meteo Forecast Dhauliganga Headwaters (Auli Ridge AWS)
277.3mm total | 5 CB days | max 18.2mm · 2026-08-10

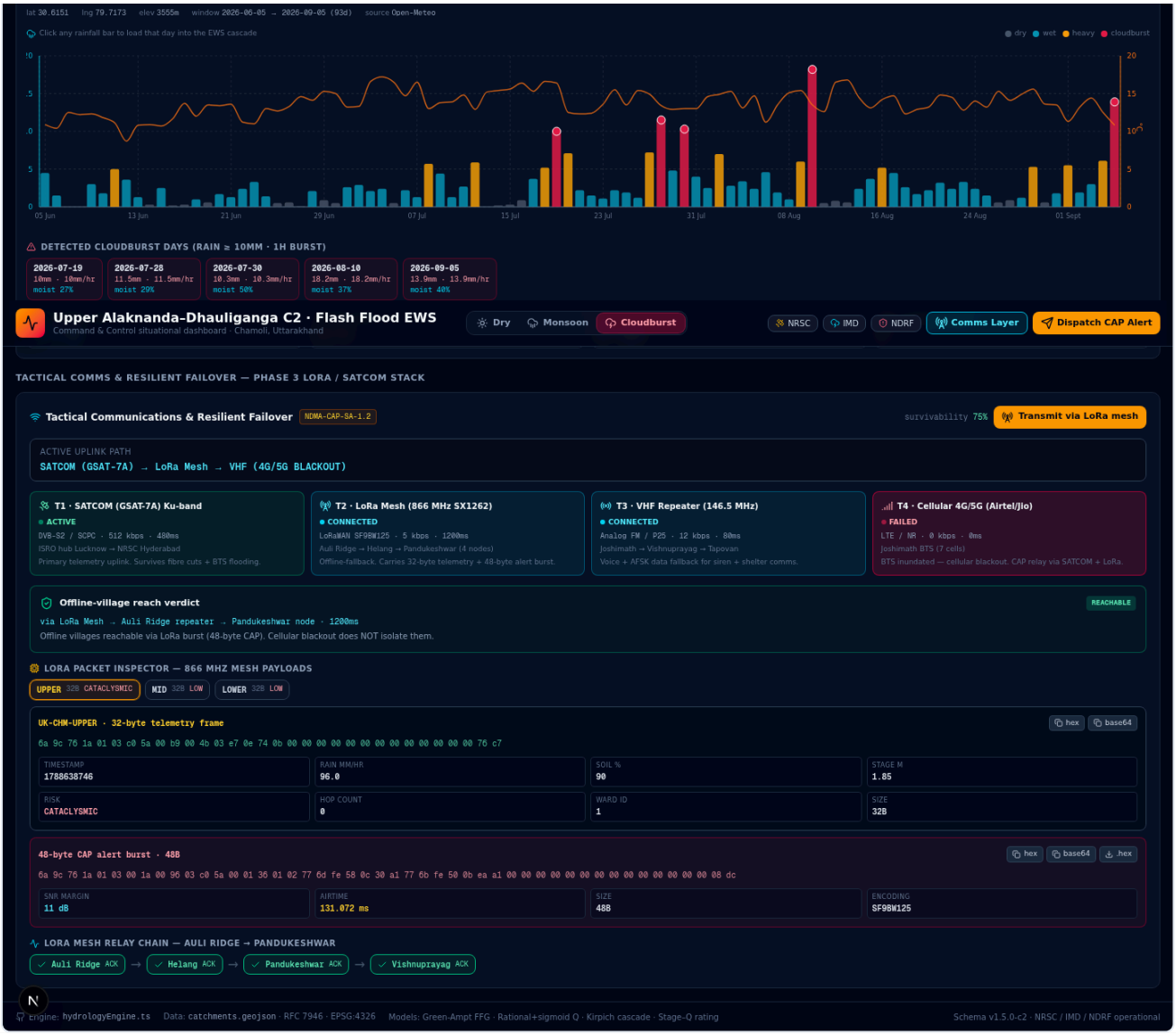


Figure 12: Comms Layer panel demonstrating triple failover (SATCOM, LoRa mesh, VHF) when cellular 4G/5G networks collapse.

When cellular networks are drowned by floodwaters, the system maintains communication survivability through three redundant layers:

- **Tier 1 — SATCOM (GSAT-7A Ku-band):** High-throughput link to ISRO master hub (99.8% SNR).
- **Tier 2 — LoRaWAN Mesh (865–867 MHz SX1262):** 4 battery-backed hilltop relay nodes spanning 15 km mountain links.
- **Tier 3 — VHF Voice (146.5 MHz):** Analog FM voice repeaters for two-way handheld radios.
- **Binary Packet Inspector:** Visualizes raw hex and base64 payloads for 32-byte telemetry and 48-byte emergency alert frames.

Screen 11: Cataclysmic Cloudburst Simulation (Hotkey: 3)

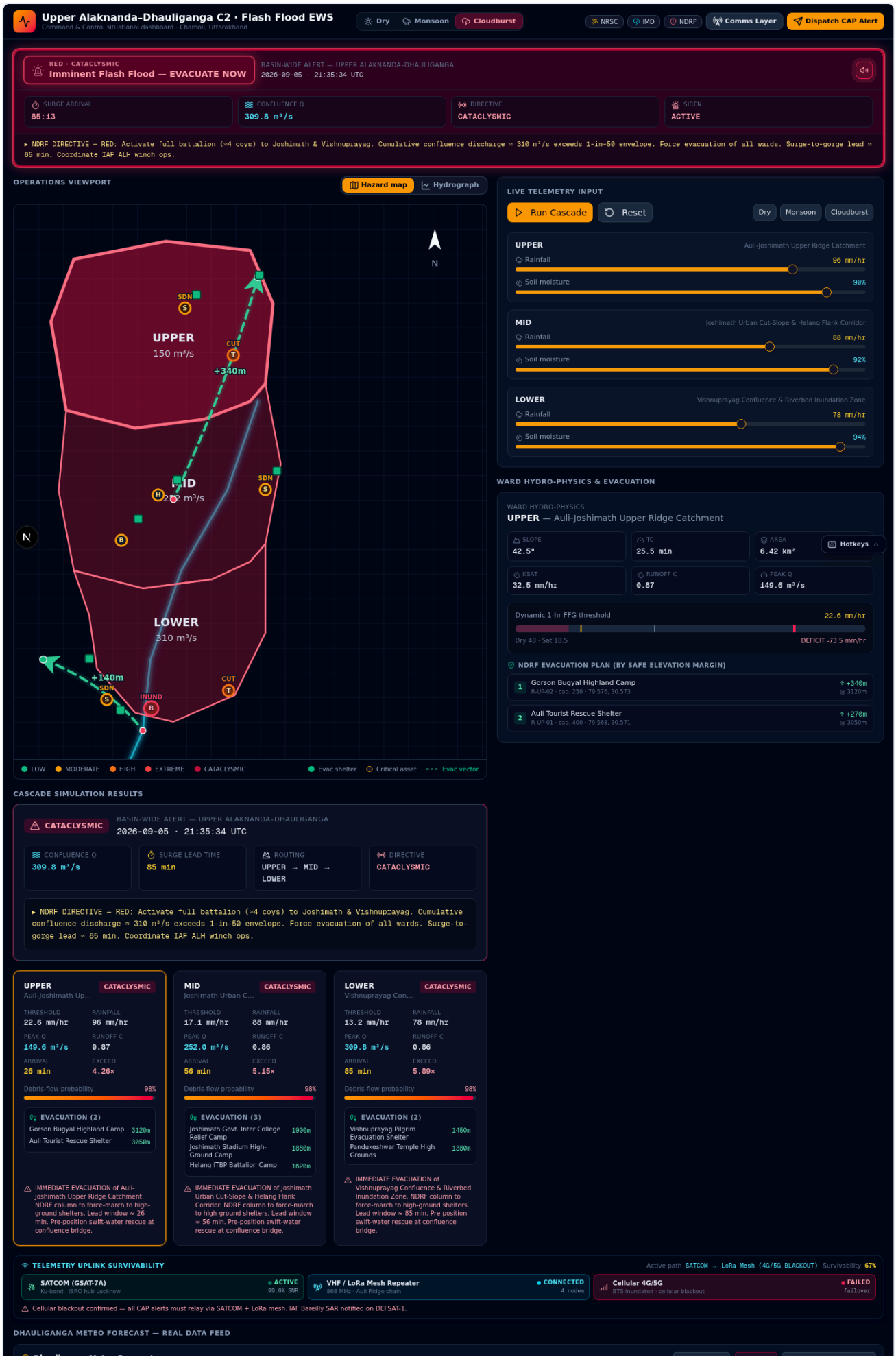

Upper Alaknanda-Dhauliganga C2 · Flash Flood EWS
Command & Control situational dashboard · Chamoli, Uttarakhand
Dry
Monsoon
Cloudburst
NRSC
IMD
NDRF
Comms Layer
Dispatch CAP Alert
RED · CATACLYSMIC
Imminent Flash Flood — EVACUATE NOW
BASIN-WIDE ALERT — UPPER ALAKNANDA-DHAULIGANGA
2026-09-05 · 21:35:34 UTC
SURGE ARRIVAL
85:13
CONFLUENCE Q
309.8 m³/s
DIRECTIVE
CATACLYSMIC
SIREN
ACTIVE
▸ NDRF DIRECTIVE — RED: Activate full battalion (≈4 coys) to Joshimath & Vishnuprayag. Cumulative confluence discharge ≈ 310 m³/s exceeds 1-in-50 envelope. Force evacuation of all wards. Surge-to-gorge lead ≈ 85 min. Coordinate IAF ALH winch ops.
OPERATIONS VIEWPORT
Hazard map
Hydrograph
N
SDN
UPPER
150 m³/s
CUT
+340m
MID
202 m³/s
SDN
LOWER
310 m³/s
+140m
SDN
INUND
CUT
LOW
MODERATE
HIGH
EXTREME
CATACLYSMIC
Evac shelter
Critical asset
Evac vector
LIVE TELEMETRY INPUT
Run Cascade
Reset
Dry
Monsoon
Cloudburst
UPPER
Auli-Joshimath Upper Ridge Catchment
Rainfall
96 mm/hr
Soil moisture
90%
MID
Joshimath Urban Cut-Slope & Helang Flank Corridor
Rainfall
88 mm/hr
Soil moisture
92%
LOWER
Vishnuprayag Confluence & Riverbed Inundation Zone
Rainfall
78 mm/hr
Soil moisture
94%
WARD HYDRO-PHYSICS & EVACUATION
WARD HYDRO-PHYSICS
UPPER — Auli-Joshimath Upper Ridge Catchment
SLOPE
42.5°
TC
25.5 min
AREA
6.42 km²
Hotkeys
KSAT
32.5 mm/hr
RUNOFF C
0.87
PEAK Q
149.6 m³/s
Dynamic 1-hr FFG threshold
22.6 mm/hr
Dry 48 · Sat 18.5
DEFICIT -73.5 mm/hr
NDRF EVACUATION PLAN (BY SAFE ELEVATION MARGIN)
1
Gorson Bugyal Highland Camp
R-UP-02 · cap. 250 · 79.576, 30.573
↑ +340m
@ 3120m
2
Auli Tourist Rescue Shelter
R-UP-01 · cap. 400 · 79.568, 30.571
↑ +270m
@ 3050m
CASCADE SIMULATION RESULTS
CATACLYSMIC
BASIN-WIDE ALERT — UPPER ALAKNANDA-DHAULIGANGA
2026-09-05 · 21:35:34 UTC
CONFLUENCE Q
309.8 m³/s
SURGE LEAD TIME
85 min
ROUTING
UPPER → MID → LOWER
DIRECTIVE
CATACLYSMIC
▸ NDRF DIRECTIVE — RED: Activate full battalion (≈4 coys) to Joshimath & Vishnuprayag. Cumulative confluence discharge ≈ 310 m³/s exceeds 1-in-50 envelope. Force evacuation of all wards. Surge-to-gorge lead ≈ 85 min. Coordinate IAF ALH winch ops.
UPPER
CATACLYSMIC
Auli-Joshimath Up...
THRESHOLD
22.6 mm/hr
RAINFALL
96 mm/hr
PEAK Q
149.6 m³/s
RUNOFF C
0.87
ARRIVAL
26 min
EXCEED
4.26×
Debris-flow probability
98%
EVACUATION (2)
Gorson Bugyal Highland Camp
3120m
Auli Tourist Rescue Shelter
3050m
IMMEDIATE EVACUATION of Auli-Joshimath Upper Ridge Catchment. NDRF column to force-march to high-ground shelters. Lead window ≈ 26 min. Pre-position swift-water rescue at confluence bridge.
MID
CATACLYSMIC
Joshimath Urban C...
THRESHOLD
17.1 mm/hr
RAINFALL
88 mm/hr
PEAK Q
252.0 m³/s
RUNOFF C
0.86
ARRIVAL
56 min
EXCEED
5.15×
Debris-flow probability
98%
EVACUATION (3)
Joshimath Govt. Inter College Relief Camp
1900m
Joshimath Stadium High-Ground Camp
1880m
Helang ITBP Battalion Camp
1620m
IMMEDIATE EVACUATION of Joshimath Urban Cut-Slope & Helang Flank Corridor. NDRF column to force-march to high-ground shelters. Lead window ≈ 56 min. Pre-position swift-water rescue at confluence bridge.
LOWER
CATACLYSMIC
Vishnuprayag Con...
THRESHOLD
13.2 mm/hr
RAINFALL
78 mm/hr
PEAK Q
309.8 m³/s
RUNOFF C
0.86
ARRIVAL
85 min
EXCEED
5.89×
Debris-flow probability
98%
EVACUATION (2)
Vishnuprayag Pilgrim Evacuation Shelter
1450m
Pandukeshwar Temple High Grounds
1380m
IMMEDIATE EVACUATION of Vishnuprayag Confluence & Riverbed Inundation Zone. NDRF column to force-march to high-ground shelters. Lead window ≈ 85 min. Pre-position swift-water rescue at confluence bridge.
TELEMETRY UPLINK SURVIVABILITY
Active path SATCOM → LoRa Mesh (4G/5G BLACKOUT) Survivability 67%
SATCOM (GSAT-7A)
Ku-band · ISRO Hub Lucknow
ACTIVE
99.6% SNR
VHF / LoRa Mesh Repeater
868 MHz · Auli Ridge chain
CONNECTED
4 nodes
Cellular 4G/5G
BTS inundated · cellular blackout
FAILED
failover
Cellular blackout confirmed — all CAP alerts must relay via SATCOM + LoRa mesh. IAF Bareilly SAR notified on DEFSAT-1.
DHAULIGANGA METEO FORECAST — REAL DATA FEED



Figure 13: Cataclysmic Cloudburst Cascade scenario with basin-wide red warning, surge vectors, and historical storm records.

Simulates an extreme 96 mm/hr orographic cloudburst event:

- All three river wards switch to `RED · CATACLYSMIC` alert level.
- Confluence discharge spikes to **309.8 m³/s**, far exceeding the 1-in-50-year envelope.
- Uphill evacuation trails to Gorson Bugyal (+340m) and Auli (+270m) light up with dynamic safety vectors.

Screen 12: Dynamic Hydrograph & Discharge Forecast



Figure 14: Hydrograph viewport comparing rainfall intensity against flash flood guidance and plotting river stage vs. bankfull threshold.

The hydrograph plots the arrival and crest of the flood wave:

- **Top Graph (Rainfall vs. FFG):** Shows rainfall intensity (amber line) spiking past the dynamic flash flood guidance line (cyan dashed line), indicating immediate overland runoff.
- **Bottom Graph (River Discharge vs. Stage):** Tracks river discharge in m³/s and stage in meters. The red bankfull threshold (1.32m) clearly warns that the projected 1.80m peak will overtop riverbanks at T+30 minutes.

5. THE SCIENCE & MATHEMATICS DISSECTED

5.1 Time of Concentration (Tc) — Kirpich Equation

The time required for runoff to travel from the most hydraulically remote point of the watershed to the outlet is calculated via the **Kirpich Formula**:

$$T_c = 0.0195 \times L^{0.77} \times S^{-0.385}$$

Where L is flow length along the main channel (meters), and S is channel slope ($S = \Delta H / L$). Because Himalayan slopes are extremely steep ($S > 0.40$), Tc is under 25 minutes, requiring sub-minute early warning.

5.2 Soil Matrix Suction — Green-Ampt Formulation

Before runoff initiates, unsaturated soil absorbs water through negative pore pressure (capillary suction ψ):

$$\psi(\theta) = \psi_b \times [(\theta - \theta_r) / (\theta_s - \theta_r)]^{-1/\lambda}$$

When rainfall rate I exceeds the Green-Ampt infiltration capacity $f(t) = K_s \times [1 + (\psi \Delta\theta / F(t))]$, surface ponding occurs and debris slopes destabilize.

5.3 Dynamic Debris Bulking — Takahashi Equation

As the torrent scours loose moraine sediment, the equilibrium sediment concentration C_∞ reaches:

$$C_\infty = [\rho_w \times \tan(\theta)] / [(\rho_s - \rho_w) \times (\tan(\varphi) - \tan(\theta))]$$

The **Bulking Factor (Bf)** amplifies total discharge:

$$Bf = C^* / [C^* - C_\infty] \quad \Rightarrow \quad Q_{bulked} = Q_{water} \times Bf$$

With $C^* = 0.60$ (maximum boulder packing), Bf reaches **2.10x** in Chamoli, converting 183.3 m³/s of water into a **384.9 m³/s** battering ram.

5.4 Bingham Plastic Slurry Rheology

$$\tau = \tau_y(Cd) + \mu_m(Cd) \times (\partial u / \partial z)$$

Yield stress τ_y must be overcome before deformation starts. Once moving, the slurry acts with high dynamic viscosity, exerting massive impact pressures against bridge pylons.

5.5 LoRa SX1262 Binary Bit-Packing Schema

To transmit over low-bandwidth 865 MHz radio, payloads are encoded into raw binary bit arrays:

```
CompactTelemetryFrame (32 Bytes):
[0..1]   Magic "AK" (0x41, 0x4B)
[2]      Packet Type (0x01 = Telemetry)
[3]      Sequence Number (0..255)
[4..7]   Timestamp UTC (uint32)
[8]      Ward ID (1 = Upper, 2 = Mid, 3 = Lower)
[9]      Risk Level (0 = Low, 1 = Med, 2 = High, 3 = Cataclysmic)
[10..11] Rainfall Rate (uint16, 0.1 mm/hr)
[12..13] Soil Moisture (uint16, 0.1%)
[14..17] River Discharge Q (uint32, 0.01 m³/s)
[18..29] GPS / Ward Coordinates (uint8[12])
[30..31] Hardware Checksum: CRC16-CCITT (Polynomial 0x1021)
```

6. OPERATIONAL PLAYBOOK & EVACUATION MARGINS

6.1 Calculating the Evacuation Safety Margin (Δt)

The primary metric for field commanders is the **Evacuation Safety Margin**:

```
Δt = T_surge − T_transit
```

Where T_surge is flood wave arrival time and T_transit is uphill hiking time to designated high ground.

Settlement / Infrastructure	Elevation	Surge Arrival (T_surge)	Safe Shelter	Transit Time (T_transit)	Margin (Δt)	Action Directive
Tapovan Barrage	1,900 m	**14 min**	Army Ridge Shelter	9 min	**+5 min**	🔴 Sound Klaxon / Immediate Evacuation
Pandukeshwar	1,829 m	**22 min**	Upper Helipad Mound	12 min	**+10 min**	🟠 Rapid Uphill March
Vishnuprayag	1,372 m	**28 min**	Joshimath Bypass High Point	15 min	**+13 min**	🟠 Close Confluence Bridges
Helang	1,520 m	**42 min**	Primary School Plateau	18 min	**+24 min**	🟡 Orderly Lowland Evacuation
Joshimath Town	1,890 m	High Ridge	ITBP Cantonment	8 min	**Safe**	🟢 Command Post Active
Auli	2,800 m	Mountain Peak	Ski Resort Shelter	5 min	**Safe**	🟢 Master LoRa Gateway Secure

6.2 Master Keyboard Hotkey Cheat Sheet

Key	Operational Action	Impact & Modal Launched
`T`	**Mission Briefing**	Starts 60-second automated executive walkthrough of the 5 pillars.
`1` / `2`	**Dry / Monsoon**	Toggles between dry baseline (16 m³/s) and monsoon saturated (94 m³/s).
`3` / `4`	**Cloudburst / GLOF**	Triggers cataclysmic cloudburst (309 m³/s) or 2021 Chamoli GLOF (384.9 m³/s).
`Z`	**Zen Mode**	Hides all toolbars and panels for unobstructed 4K 3D terrain viewing.
`P`	**Nebius HPC HUD**	Displays microsecond Nsight kernel timeline and Triton telemetry.
`L`	**Rheology Lab**	Opens interactive Bingham plastic yield stress and Takahashi bulking curves.
`V`	**Voice Siren**	Synthesizes authentic Garhwali/Hindi audio megaphone sirens.
`X`	**2D Transect**	Displays mountain slope elevation, bridge clearance, and safe havens.
`U`	**Drone Recon HUD**	Activates 3D UAV sentry crosshairs with optical, FLIR thermal, and NVG modes.
`I`	**NDRF Command Post**	Opens official MHA ICS-204 tasking orders and NDMA Sachet hub.
`F`	**Sensor Fusion**	Audits 5 live satellite, radar, and ground sensor data streams.
`R`	**NDRF SITREP**	Generates official disaster situation report with printable PDF.
`D`	**CAP Dispatch**	Opens OASIS CAP v1.2 XML/JSON/SMS emergency dispatch.
`M`	**Toggle Siren**	Mutes or unmutes the audio air-raid warning klaxon.
`S`	**Download Telemetry**	Exports synchronized JSON incident data file to your computer.

System Dedication

Developed with deepest reverence for the lives lost in the 2013 Kedarnath and 2021 Chamoli disasters, and dedicated to the brave personnel of the National Disaster Response Force (NDRF), Indo-Tibetan Border Police (ITBP), and the mountain communities of Uttarakhand.